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Research Paper

A new 2D continuous-discontinuous heat conduction model for modeling heat transfer and thermal cracking in quasi-brittle materials

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ABSTRACT

This paper proposes a new 2D continuous-discontinuous heat conduction model to simulate heat transfer and thermal cracking in fractured quasi-brittle materials by combining with mechanical calculations of the Finite-Discrete Element Method. By updating the sharing relationship of the nodes at the cracks, the model can accurately predict the temperature evolution, crack propagation and considers the effect of cracking on heat conduction. We verified the correctness of the continuous-discontinuous heat conduction model using examples of heat conduction in a continuum, media with single or multiple cracks, thermal deformation problems, and cracking induced by temperature gradient and thermal mismatch. The numerical results indicate that the continuous-discontinuous heat conduction model well reflects the effect of crack propagation on heat conduction and the discontinuity of temperature across the cracks. It provides a powerful tool for studying the whole process and mechanism analysis of thermal cracking of quasi-brittle materials.

1. Introduction

Quasi-brittle materials, such as rock, concrete, and ceramics, are widely used in modern engineering practice. Conduction of heat in these solids is an important consideration (Carslaw and Jaeger, 1959; Ezekoye, 2016), as the temperature change may alter the thermal and mechanical properties of quasi-brittle materials (Gautam et al., 2018; Heuze, 1983), and even cause the material to deteriorate or result in engineering accidents. For example, when a concrete structure is on fire, its temperature can rise rapidly to 700 °C or higher in a short time (Ulm et al., 1999). A large temperature gradient in concrete can cause thermal cracking in the structure (Committee, 2007). In CO_2 geological storage, when CO_2 is injected into the brine layer at a depth of about 1000–2500 m, the surrounding salty rock may shrink or even crack due to the low-temperature CO_2 (Taylor and Bryant, 2014). In the underground storage of nuclear waste, the heat released in nuclear decay may heat the surrounding rock and cause thermal cracking. Accidents of nuclide leakage may even happen (Kim et al., 2011). Therefore, it is of great significance to study the thermal cracking mechanism of quasi-brittle materials under the coupled thermal-mechanical effect.

A variety of experimental technologies, such as acoustic emission technology (AE) (Kumari et al., 2017; Zhao et al., 2019), scanning electron microscope (SEM) (Kumari et al., 2019; Mahanta et al., 2016; Ouedraogo et al., 2011), CT microscopy scanning technology (Kim et al., 2011), thermal stress device (TSD) (Amin et al., 2009), optical microscope (Peng et al., 2019), etc., have been used to study the thermal and mechanical properties of a quasi-brittle material (Ahmed et al., 2018; Kim et al., 2011; Sun et al., 2016). However, the experimental study is not only time-consuming and costly but also difficult to observe the initiation and propagation of micro-cracks during the thermal cracking process. Thus, numerical methods are useful to explore the mechanism of thermal cracking in quasi-brittle materials, which is the scope of the present study.

The numerical simulation methods for thermal cracking of quasi-brittle materials can be generally divided into two types: continuum-based method and discontinuous-based method. In the continuum-based methods, finite element method (FEM) (Bou Jaoude et al., 2018; Fu et al., 2017; Meguid and Hu, 1999; Nechnech et al., 2002) and its variants, such as extended finite element method (XFEM) (Areias and Belytschko, 2005; Belytschko and Black, 1999; Jiang et al., 2020; Moës et al., 2017) and particle finite element method (PFEM) (Rodriguez et al., 2016), are the most widely used to model the thermal cracking of quasi-brittle materials, followed by finite difference method (FDM) (Ekneligoda and Marshall, 2018; Najafi et al., 2014; Wang et al.,

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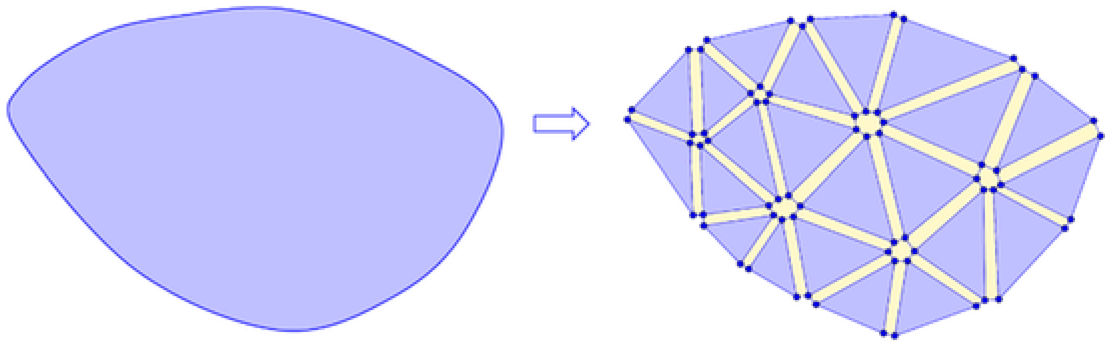

Fig. 1. The representation of the continuum in FDEM and the connection between the joint element and the triangular element.

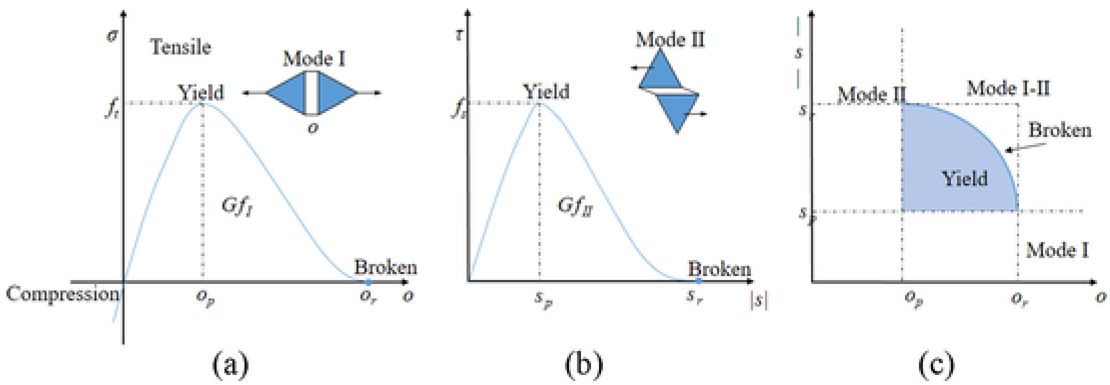

Fig. 2. Constitutive behavior of joint element: (a) Mode I, the relationship between normal stress and opening and compressing of the joint element; (b) Mode II, the relationship between shear stress and the slip of joint element; (c) Mode I-II, the relationship between the broken type and the opening and slip amounts (Lisjak et al., 2013).

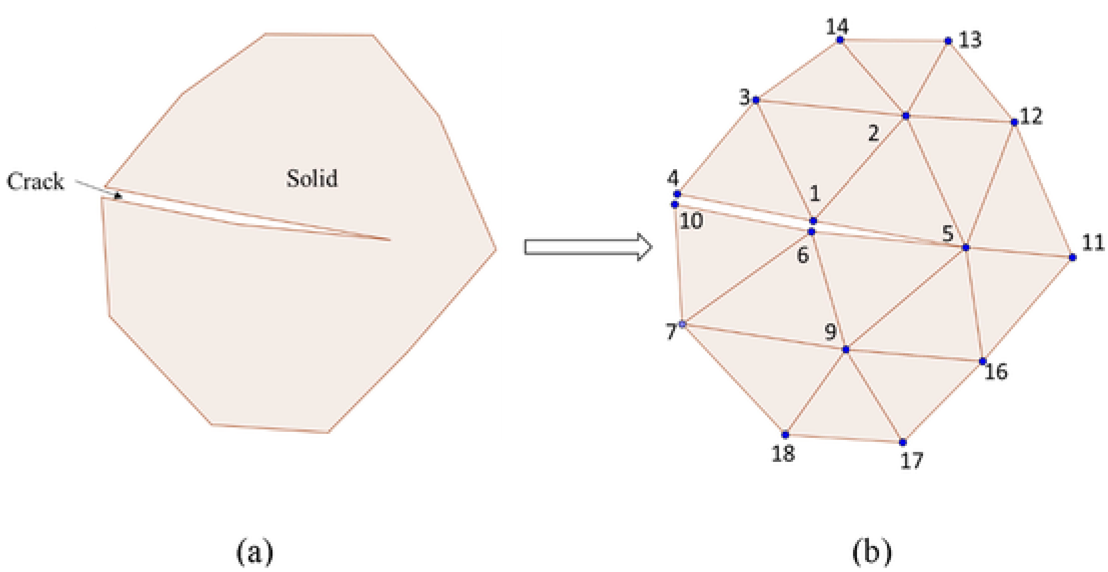

Fig. 3. Discretization of the problem domain. (a) discontinuous medium. (b) The initial mesh on which the heat transfer calculation is based.

2019a) and boundary element method (BEM) (An et al., 2018; Fang et al., 2020; Gao et al., 2015; Yang et al., 2017). In recent years, more advanced continuum-based methods, such as peridynamics (PD) (Bazazzadeh et al., 2020; Shou and Zhou, 2019; Wang and Zhou, 2019; Wang et al., 2018; Wang et al., 2019b; Wang et al., 2019c) and phase-field method (Borden et al., 2018; Chu et al., 2017; Li and Shirvan, 2021; Liu et al., 2020a; Miehe et al., 2015a; Miehe et al., 2015b), have been successfully applied to model the thermal cracking of quasi-brittle material. However, it is difficult for these methods to handle the propagation of multiple cracks, and the hindering effect of cracking on heat conduction has not been considered.

To overcome the limitations of the continuum-based method, the discontinuous-based method has been widely used to simulate the thermal cracking of quasi-brittle materials, such as the discrete element method (DEM) (Choo et al., 2012; Liu et al., 2019; Xia, 2015; Xia et al., 2014), discontinuous deformation analysis (DDA) (Jiao et al., 2015), and others (Caggiano and Etse, 2015; Cui et al., 2019; Kidane et al., 2010; Liu et al., 2020b). However, the heat conduction calculation in these methods is completely independent of the mechanical calculation.

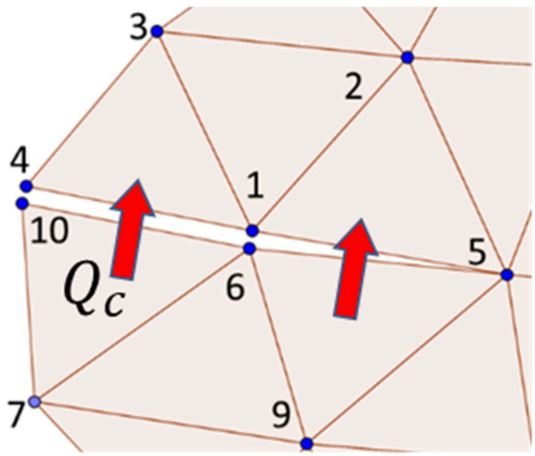

Fig. 4. Heat exchange between triangular elements Δ1,3,4 and Δ6,10,7 through a crack.

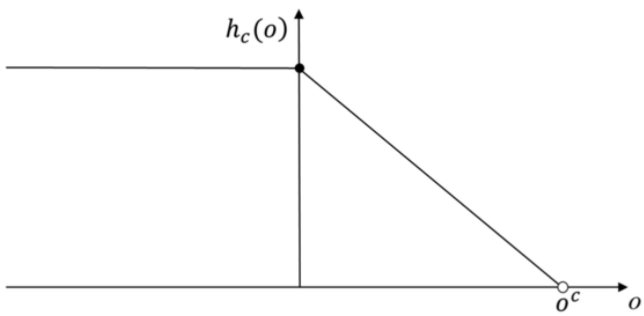

Fig. 5. A simple example of the relationship between h_c and the crack opening (o) (Benabou et al., 2013).

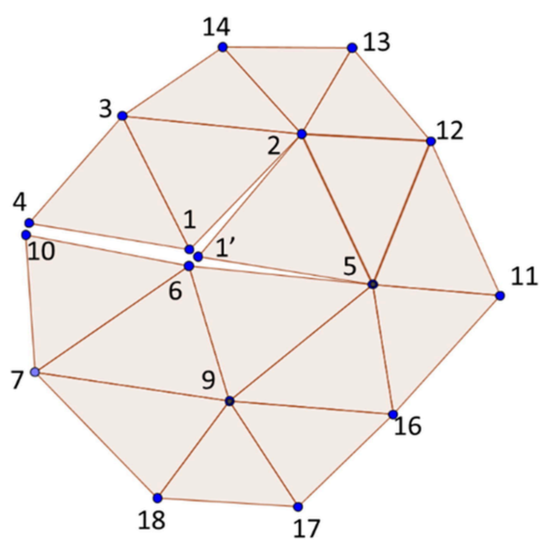

Fig. 6. Calculation of heat conduction during crack propagation.

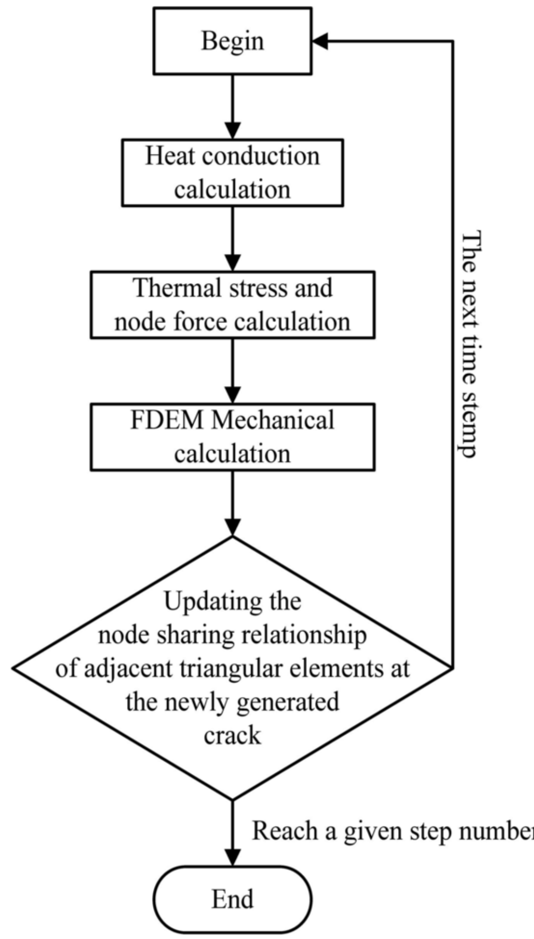

Fig. 7. Thermo-mechanical coupling calculation process.

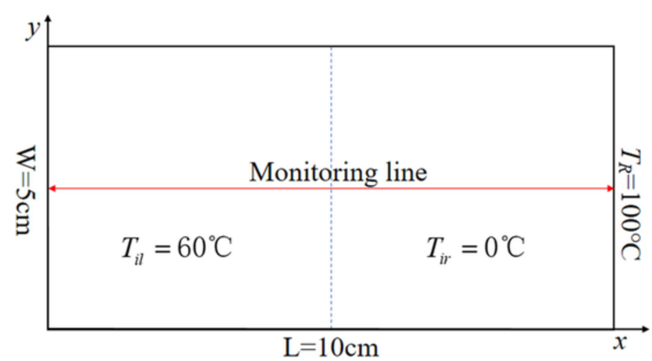

Fig.8. Schematic model and monitoring line.

Even after the cracks are generated, the calculation mesh on which the heat conduction calculation is based still shares nodes at the cracks. Therefore, the temperature on both sides of the cracks is continuous and the effect of cracking on heat conduction cannot be considered.

To simulate the transition from continuum to discontinuous, Munjiza (2004) and Munjiza et al. (1999b, 1995) proposed the combined finite discrete element method (FDEM), which is suitable for modeling the cracking of quasi-brittle material (Deng and Liu, 2020; Fukuda et al., 2019; Lei et al., 2020; Lei et al., 2019; Li et al., 2020; Li et al., 2021). FDEM uses the framework of the finite element method to calculate the stress and strain in the solid element. Thus, the concept of stress-strain in continuum mechanics is well preserved. The discrete element method is applied to treat the contact of various discontinuous surfaces in FDEM. Therefore, FDEM can dynamically deal with contact problems during block collision, crack sliding, and closure. However, the original FDEM is only capable of pure mechanical calculation. Later, Yan and Jiao (2018), Yan and Zheng (2016, 2017b, 2017c), and Yan et al. (2018) developed coupled hydro-mechanical models based on FDEM for simulating hydraulic fracturing or fluid-driven fracturing in complex fractured porous media. Yan and Zheng (2017a) also built a 2D coupled model to solve thermo-mechanical problems. However, in this model, the influence of crack on heat conduction cannot be considered because the medium is always regarded as a continuum in heat transfer calculation. Recently, Yan and Jiao (2019) and Yan et al.

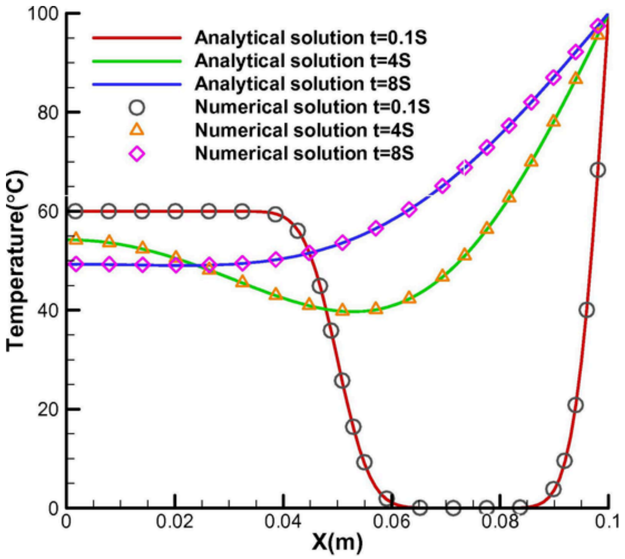

Fig. 9. The comparison of the temperatures of numerical and analytical solutions.

(2019) built a simplistic heat conduction model and thermo-mechanical model using joint elements. However, to minimize the hindrance of the joint elements to the heat transfer in the continuum, the heat exchange coefficient of the unbroken joint element needs to be sufficiently large, which requires a much reduced time step (on the order of 100 times) compared with the case when the joint elements are not used.

To overcome the above-mentioned shortcomings of the model, we propose a new 2D continuous-discontinuous heat conduction model. Calculation of heat conduction can dynamically update the node sharing relationship of adjacent triangular elements at the crack based on crack propagation. The model in this manuscript considers the influence of crack propagation on heat transfer without the need to introduce joint elements and its heat exchange coefficient. Therefore, the calculation efficiency of the model is significantly improved (more than 100 times) compared with that in the literature (Yan et al., 2019). Besides, the 2D heat conduction model can be combined with various numerical methods to construct coupled thermo-mechanical models. As an example, we combine the heat conduction model with FDEM to construct the coupled thermo-mechanical model for solving the thermal cracking problem of quasi-brittle materials.

This paper mainly consists of the following parts. In the second section, the basic principles of FDEM and fracture criterion of the joint element are briefly introduced. In the third section, the construction of the 2D continuous-discontinuous heat conduction model is introduced. In the fourth section, thermal stress calculation, the coupled thermo-mechanical model, and its calculation process are introduced. In the fifth section, an example of continuum heat conduction with an analytical solution is given to validate the model in dealing with continuum heat conduction. Then, examples of heat conduction in discontinuous media with single crack or multiple cracks are given. Next, we deduce an analytical solution to a thermal deformation problem and use it to verify the coupled thermo-mechanical model. Finally, two thermal cracking experiments are simulated by the coupled model.

2. Fundamental principles of FDEM

The principles of FDEM are that the problem domain is discretized into a triangular finite element mesh and the joint element with zero-thickness is inserted between the adjacent triangular elements. Thus, the adjacent triangular element does not share nodes, as shown in Fig. 1. The crack initiation, propagation, and intersection in the continuum are modeled by the breaking of the joint element. Before the joint element breaks, the deformation of the continuum is modeled by that of the constant strain triangular element and the joint element with the bonding effect. A detailed introduction about FDEM can be found in the papers Munjiza and Andrews (1998, 2000) and Munjiza et al. (1999a) and Munjiza and John (2002). Here we only introduce the constitutive equation of the joint element.

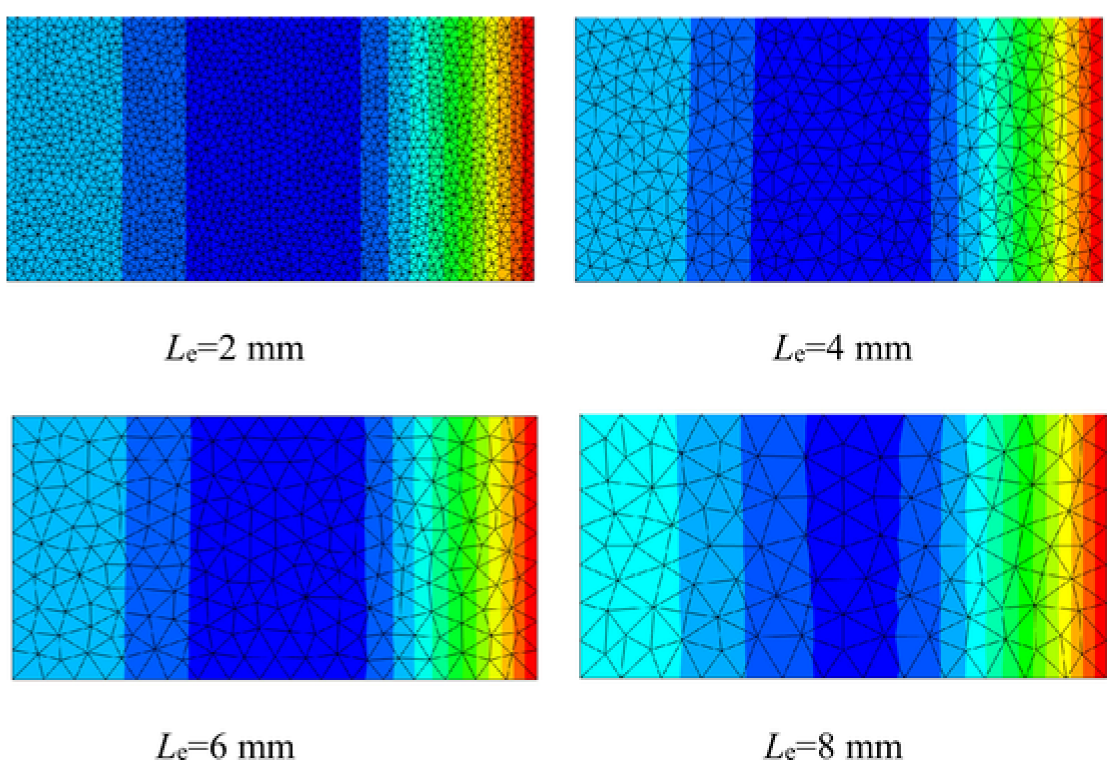

Fig. 10. Temperature distribution under different element sizes (L_e) at t = 4 s.

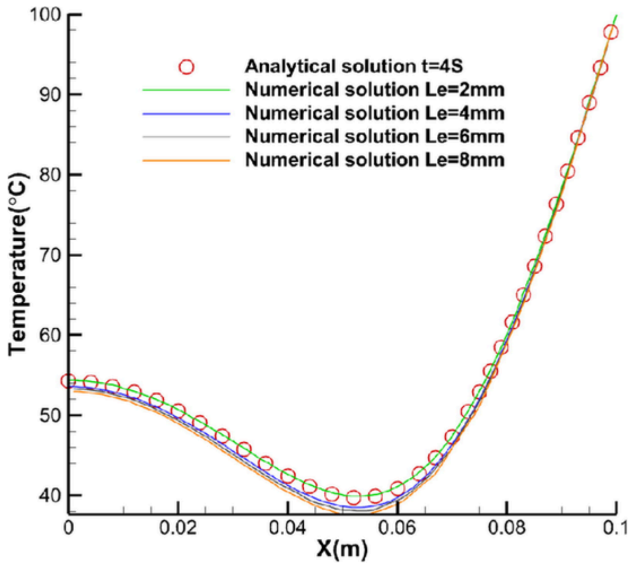

Fig. 11. Comparison of numerical results and analytical solution using different element sizes.

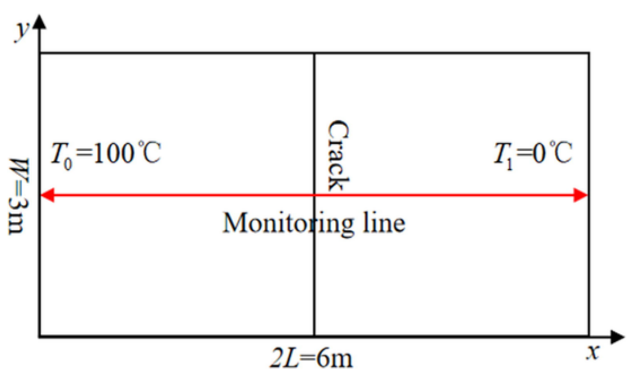

Fig. 12. Schematic model and monitoring line.

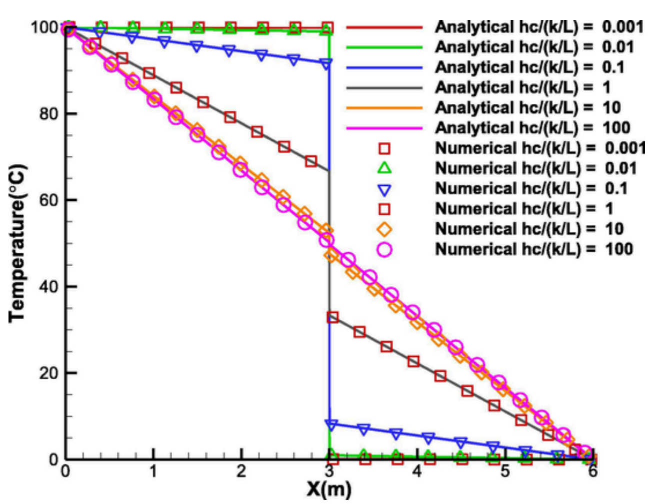

Fig. 13. Temperature distribution along x-direction at steady state under different $h_c/(k/L)$ conditions.

In FDEM, the joint element is inserted into the adjacent triangular elements. The two triangular elements connected by the joint element may move relative to each other under the action of external load. The relative motion can be decomposed into the displacement vertical and parallel to the joint plane, that is, the normal displacement o and the tangential displacement s. The normal displacement is positive for

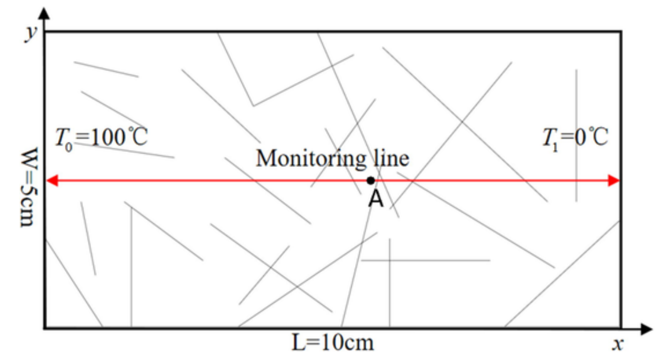

Fig. 14. . Schematic model and monitoring line.

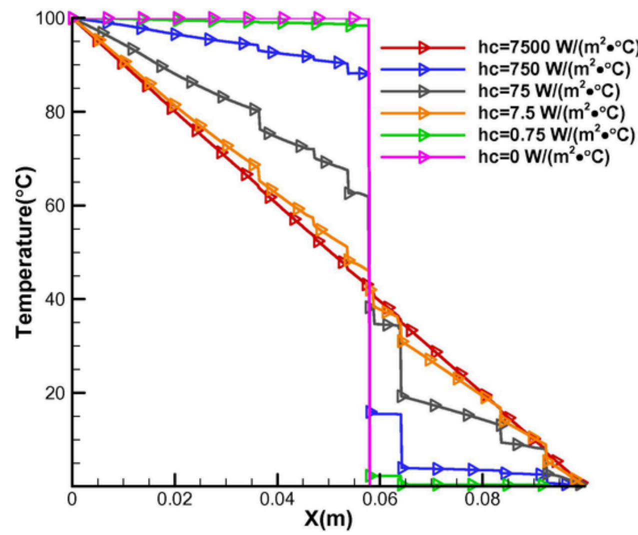

Fig. 15. Temperature distribution along the monitoring line under different heat exchange coefficients of the crack.

opening and negative for closing. According to the magnitude of the normal displacement o and the tangential displacement s, the damage state of the joint element can be determined. In the case of pure tension, the normal displacement is equal to the critical normal opening o_p when the normal stress of the joint element reaches the tensile strength; in the case of pure shear, the tangential displacement is equal to the critical tangential displacement s_p when the tangential stress of the joint element reaches the shear strength. According to the relationship between the normal displacement o and the tangential displacement s and the critical normal and tangential displacement o_p, s_p, the damage variable D of the joint element is defined by

$$D=\begin{cases} 0, & o<o_p \text{ and } s<s_p \\ \frac{o-o_p}{o_r-o_p}, & o\geqslant o_p \text{ and } s<s_p \\ \frac{s-s_p}{s_r-s_p}, & o<o_p \text{ and } s\geqslant s_p \\ \sqrt{\left(\frac{o-o_p}{o_r-o_p}\right)^2+\left(\frac{s-s_p}{s_r-s_p}\right)^2}, & o\geqslant o_p \text{ and } s\geqslant s_p \end{cases} \quad (1)$$

where o_r is the normal displacement of the joint element when the joint element occurs pure tensile failure; s_r is the tangential displacement of the joint element when the joint element occurs pure shear failure. If the damage variable $D > 1$ calculated according to Eq. (1), then set $D = 1$.

According to the damage variable D of the joint element, we can obtain the reduction coefficient $f(D)$ of the normal and tangential stress of the joint element by

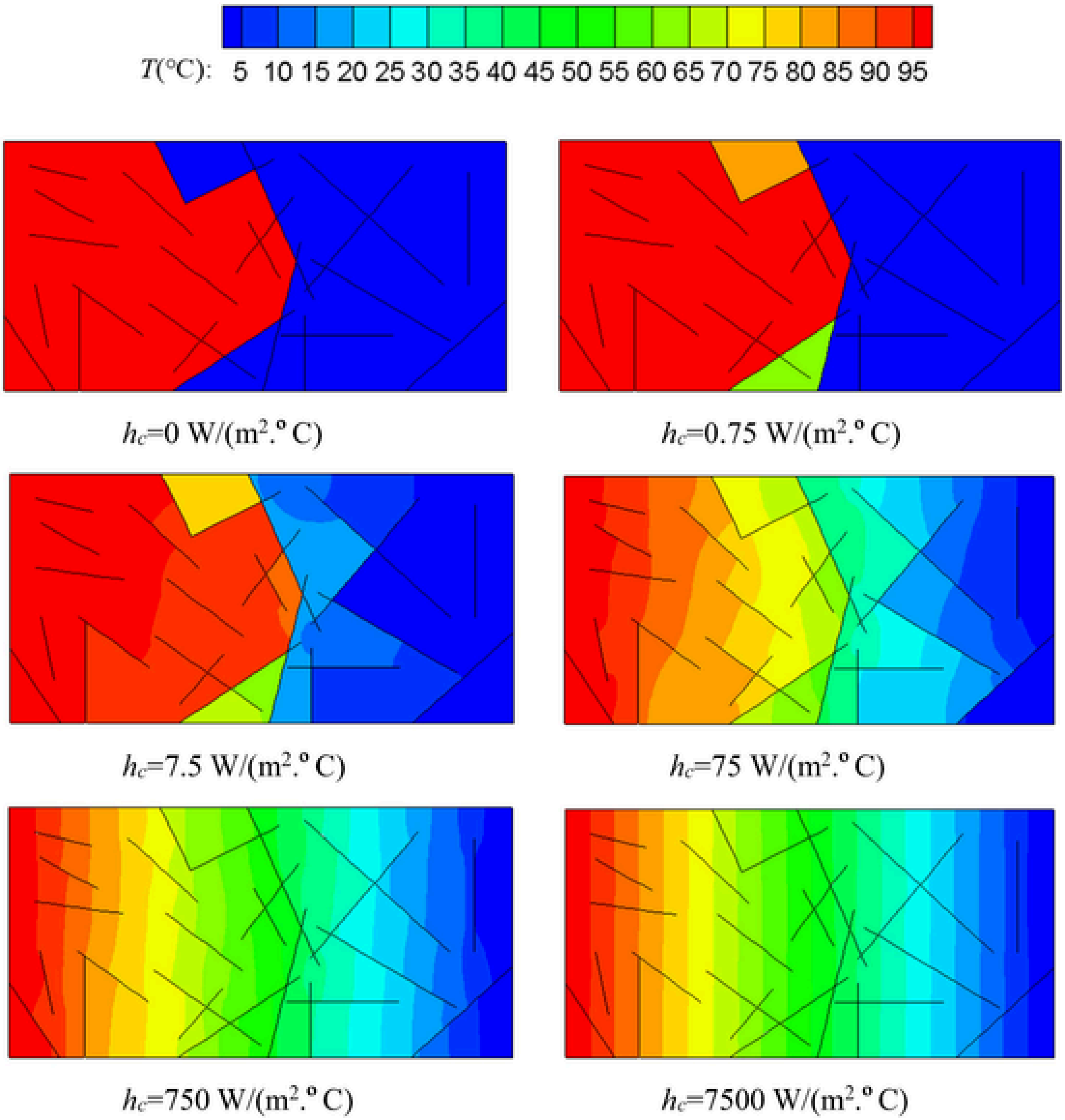

Fig. 16. Temperature distribution along the monitoring line under different heat exchange coefficients of the crack.

$$f(D) = \left(1 - \frac{a+b-1}{a+b} e^{\left(\frac{D(a+cb)}{(a+b)(1-a-b)}\right)}\right) \left(a(1-D) + b(1-D)^{c}\right) \tag{2}$$

where a, b and c are the parameters obtained by fitting the experiment curve when the material is under tension, where $a = 0.63$, $b = 1.8$ and $c = 6.0$.

In this way, the normal stress and tangential stress of the joint element can be expressed as

$$\sigma = \begin{cases} \frac{2o}{o_p} f_t \text{ if } o<0 \\ \left(\left[\frac{2o}{o_p} - \left(\frac{o}{o_p}\right)^2\right] f_t\right) f(D) \text{ if } 0 \leqslant o \leqslant o_p \\ f(D) f_t \text{ if } o>o_p \end{cases} \tag{3}$$

$$\tau^{coh} = \begin{cases} \left[\frac{2|s|}{s_p} - \left(\frac{|s|}{s_p}\right)^2\right] (-\sigma \tan(\phi) + c) \ f(D) \text{ if } 0 \leqslant |s| \leqslant s_p \\ (-\sigma \tan(\phi) + c) \ f(D) \text{ if } |s| > s_p \end{cases} \tag{4}$$

If the joint element is in a pure tension state or pure shear state, Eqs. (3) and (4) can be visually shown in Fig. 2 (a) and (b), respectively. If the joint element is in the state of mixed tension shear state, whether the joint element is in a yield or failure state can be visually shown in Fig. 2(c).

3. A 2D continuous-discontinuous heat conduction model considering crack propagation

Based on the mesh shown in Fig. 3, in the continuous-discontinuous heat conduction model, the entire problem domain is discretized into many triangular elements. The adjacent triangular elements on two sides of the crack do not share nodes while the other adjacent triangular elements share nodes. Thus, the mesh used for the heat conduction calculation is different from that of FDEM mechanical calculation. The temperature distribution of a point in a triangular element can be obtained by the temperature of its three nodes according to linear interpolation. The heat conduction in the triangular element is described by Fourier's law. Besides, heat exchange may also occur between the adjacent triangular elements through the cracks, which can be determined by the temperature difference between the two sides of the crack and the heat exchange coefficient of the crack. Finally, we can calculate the temperature distribution of the entire problem domain.

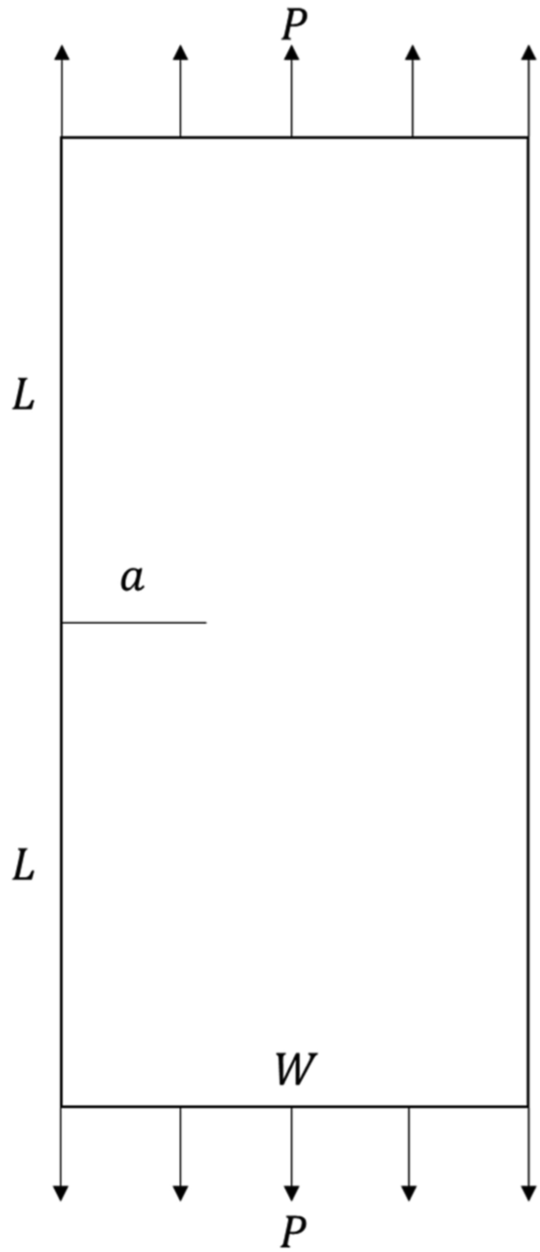

Fig. 17. A single edge notched plate under tension.

3.1. 2D continuous-discontinuous heat conduction model

Based on Fourier's law in heat conduction, the heat flux per unit area in i-th direction q_i is given by

$$q_i = -k_{ij}\frac{\partial T}{\partial x_j} \tag{5}$$

where k_{ij} is the thermal conductivity tensor, T is the temperature and x_j is the position vector.

For any given mass M, the temperature change can be expressed as

$$\frac{\partial T}{\partial t} = \frac{Q_{total}}{C_p M} \tag{6}$$

where Q_{total} is the total heat flux into the mass M, C_p is the specific heat, $t$is the time.

The discontinuity (see Fig. 3 (a)) is discretized into many triangular elements to form the initial mesh for the calculation of continuous-

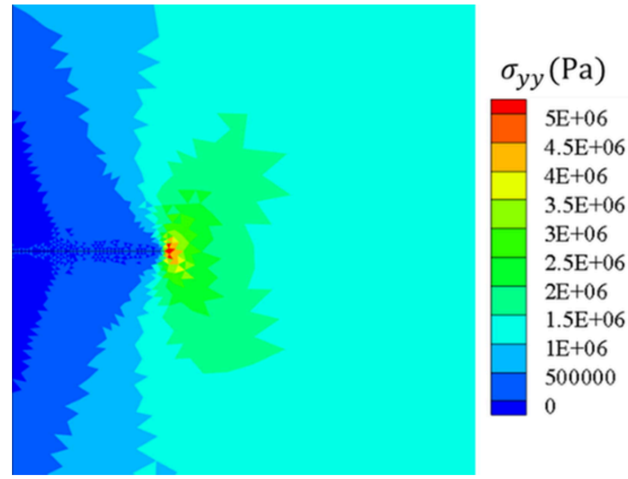

Fig. 18. Vertical stress σ_{yy} distribution at the crack tip.

discontinuous heat conduction, as shown in Fig. 3 (b). The adjacent triangular elements on both sides of the crack do not share nodes. According to the temperature of the nodes of the triangular element, the temperature of a point within the triangular element can be calculated by linear interpolation. We evenly distribute the mass of the triangular element, i.e., 1/3 of the element mass to its three nodes. Then, the temperature distribution of the entire problem domain is converted to the temperature of a finite number of nodes of triangular elements. Finally, according to the topological connection between the elements and nodes shown in Fig. 3 (b), the calculation of the temperature evolution of the problem domain is introduced.

As shown in Fig. 3 (b), heat conduction may occur in Δ1,2,3, Δ1,3,4 and Δ1,5,2 that connect to node 1 because the temperature of node 1 may be different from that of nodes 2, 3, 4, and 5. Taking the triangular element Δ1,2,3 as an example, the temperature of a point in the triangle follows a linear distribution. Then, the temperature gradient in the triangular element is constant and is given by

$$\frac{\partial T}{\partial x_i} = \frac{1}{A}\int_A \frac{\partial T}{\partial x_i} dA \tag{7}$$

According to Gauss's theorem, Eq. (7) can be written as

$$\frac{\partial T}{\partial x_i} = \frac{1}{A}\int_s T n_i ds = \frac{1}{A}\sum_{m=1}^{3} \overline{T}^m \in_{ij} \Delta x_j^m \tag{8}$$

where A is the area of the triangular element, n_i is the outward normal unit vector, $\overline{T}^m$ is the average temperature of the edge m, Δx^m_j is the difference in x_j between two nodes of the edge m, and $\in_{ij}$ is the 2D permutation tensor, $\in = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Substituting Eq. (8) into Eq. (5), the heat flux per unit area in the x-axis and y-axis can be calculated as q_x and q_y. Therefore, the heat flux into node 1 through the triangular element Δ1,2,3 can be calculated by

$$Q_{\Delta 1,2,3\rightarrow 1} = \frac{q_i n_i^{(1)} L^{(1)}}{2} \tag{9}$$

where, $n_i^{(1)}$ is the outward normal unit vector of the edge opposite to node 1, $L^{(1)}$ is the length of the edge opposite to node 1 within Δ1,2,3. Similarly, we can obtain the heat flux $Q_{\Delta 1,3,4\rightarrow 1}$ and $Q_{\Delta 1,5,2\rightarrow 1}$ from the triangular elements Δ1,3,4 and Δ1,5,2 into node 1, respectively.

Heat conduction can also occur between elements on both sides of a crack. For example, the triangular elements on the other side of the crack, as shown in Fig. 4, Δ6,10,7, and Δ6,9,5 exchange heat to Δ1,3,4 and Δ1,5,2 through the crack. Taking the triangular elements Δ6,10,7

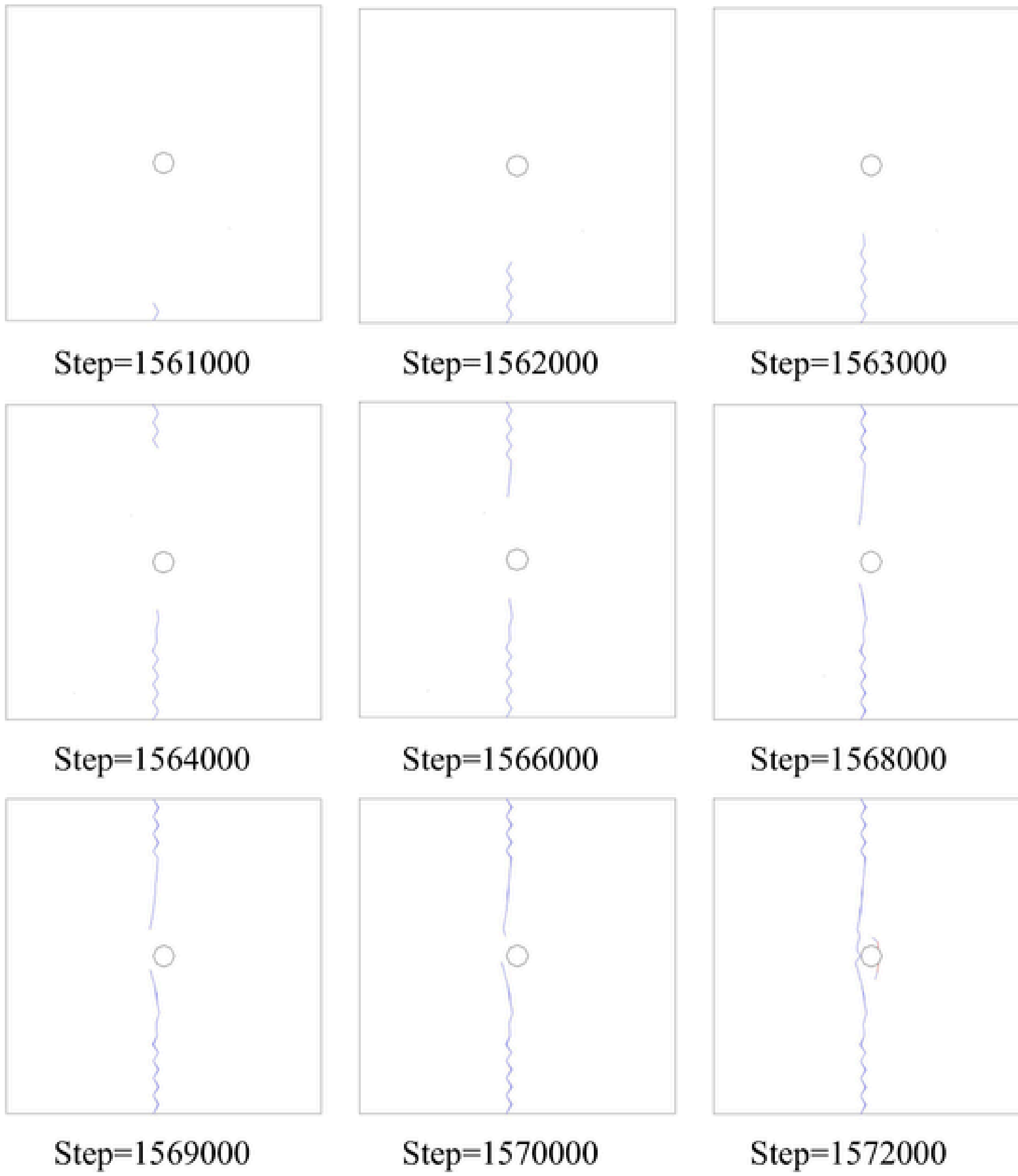

Fig. 19. The crack initiation and propagation at different time steps.

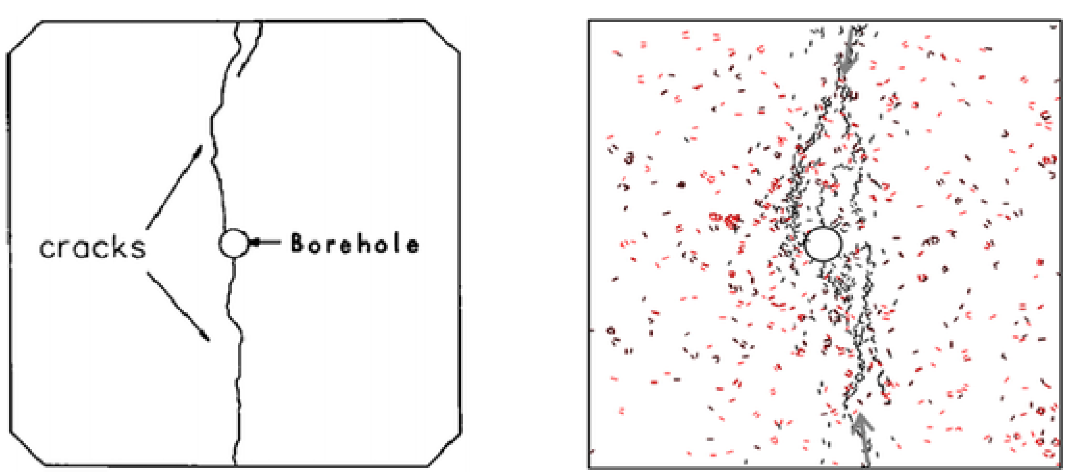

Fig. 20. (a) Laboratory experiment result (Jansen et al., 1993), (b) PFC simulation result (Zhao, 2015).

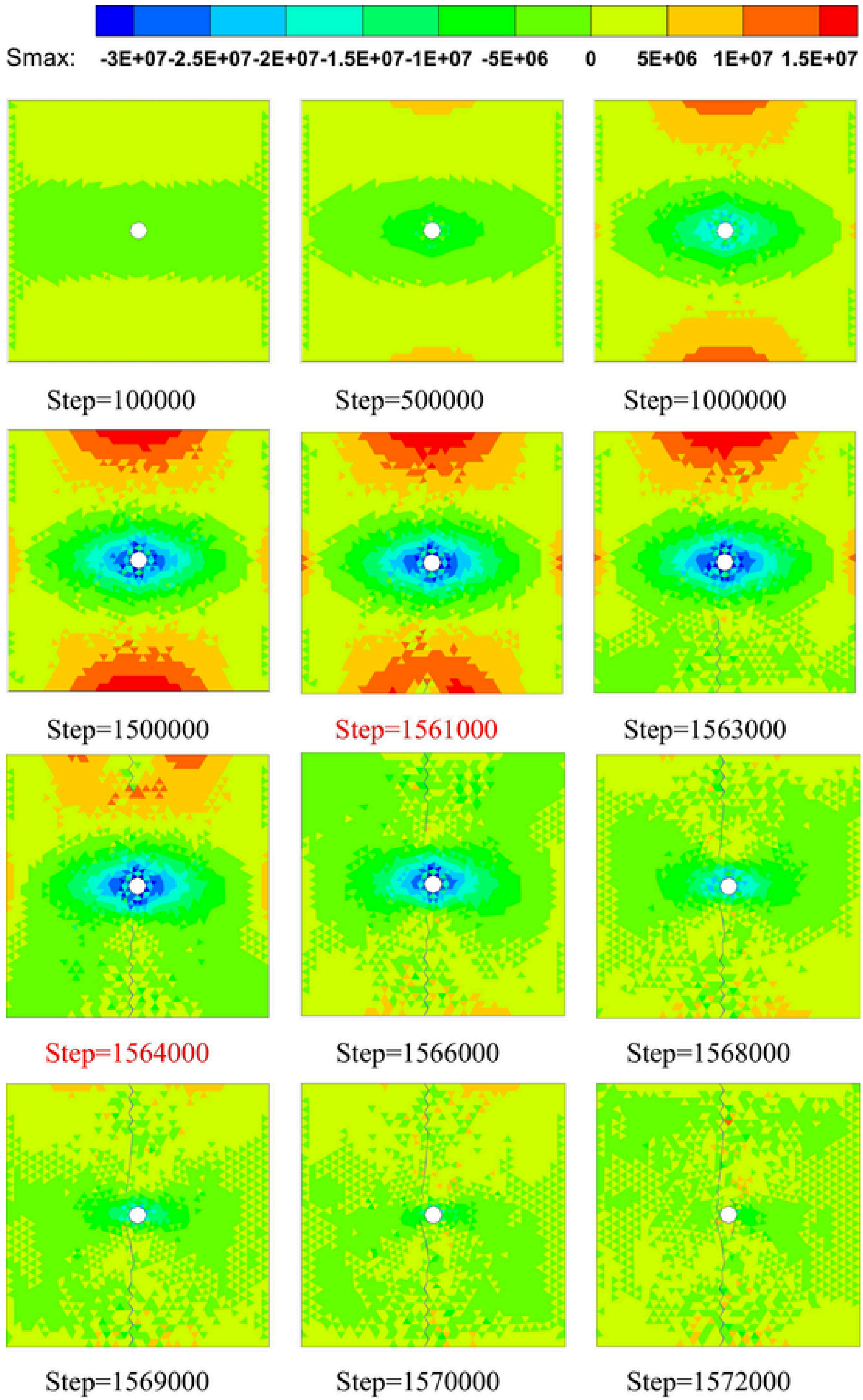

Fig. 21. The evolution of maximum principal stress at different time steps.

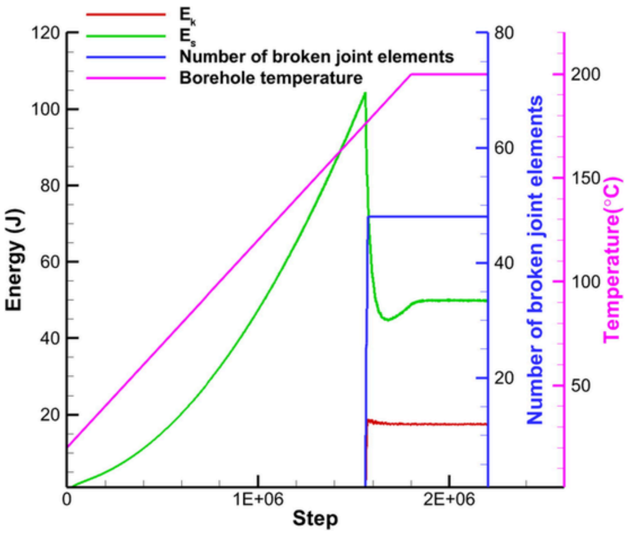

Fig. 22. Evolution of kinetic energy (E_k), strain energy (E_s), and the number of broken joint elements during the heat conduction and thermal cracking process.

Table 1
Material parameters.

Minerals	Matrix	Inclusion
Tensile strength/(MPa)	14	50
Modulus of elasticity/(GPa)	30	80
Poisson's ratio	0.16	0.22
Thermal expansion coefficient/(10^{-6}/°C)	9	14

and Δ1,3,4 as an example, the temperature of nodes 1, 4, 6, and 10 are T_1, T_4, T_6, and T_{10}, respectively. Then, the heat flux exchanged from the triangular elements Δ6,10,7 to Δ1,3,4 through the crack is given by

$$Q_c = \frac{1}{2}h_c(T_{10} + T_6 - T_4 - T_1)L_{6,10} = \frac{1}{2}h_c(T_{10} - T_4)L_{6,10} + \frac{1}{2}h_c(T_6 - T_1)L_{6,10} \tag{10}$$

where, h_c is the heat exchange coefficient of the crack. It should be pointed out that even if the crack is in an open state, h_c is not 0. Because the triangular elements on both sides of the crack can still transfer heat by heat radiation and air convection. But as the crack opening increases, the heat transfer between the triangular elements on both sides of the crack gradually decreases. In order to consider this effect, we express h_c as a function of crack opening (o), as shown in Fig. 5 (Benabou et al., 2013). It needs to be pointed out that this is only a simple example of the relationship between h_c and the crack opening (o). Other functional form can be used based experimental measurement.

In the above equation, the heat flux assigned to node 1 from Δ6,10,7 into Δ1,3,4 through the crack is given by

$$Q_{\Delta 6,10,7\to 1} = \frac{1}{2}h_c(T_6 - T_1)L_{6,10} \tag{11}$$

Similar to Eq. (11), the heat flux assigned to node 1 from the triangular element Δ6,9,5 into Δ1,5,2 through the crack is given by

$$Q_{\Delta 6,9,5\to 1} = \frac{1}{2}h_c(T_6 - T_1)L_{5,6} \tag{12}$$

Therefore, the total heat flux into node 1 can be written as

$$Q_{total\to 1} = Q_{\Delta 1,2,3\to 1} + Q_{\Delta 1,3,4\to 1} + Q_{\Delta 1,5,2\to 1} + Q_{\Delta 6,9,5\to 1} + Q_{\Delta 6,10,7\to 1} \tag{13}$$

According to Eq. (6), the temperature of node 1 at the next time step is

$$T_1^{t+\Delta t} = T_1^t + \frac{Q_{total\to 1}}{C_p M}\Delta t \tag{14}$$

Similarly, the temperature of other nodes on the crack, node 4, 5, 6, 10, can be updated. For the nodes that are not located on the crack, the total heat flux into the node is equal to the sum of heat flux into the nodes from all the neighboring triangular elements without considering heat flux through the crack. For example, the total heat flux into node 2 is given by

$$Q_{total\to 2} = Q_{\Delta 1,2,3\to 2} + Q_{\Delta 1,5,2\to 2} + Q_{\Delta 5,12,2\to 2} + Q_{\Delta 2,13,12\to 2} + Q_{\Delta 2,13,14\to 2} + + Q_{\Delta 2,14,3\to 2} \tag{15}$$

Finally, we can obtain the temperature evolution of the entire problem domain. For the model proposed in this paper, the adjacent triangular elements in the continuum share nodes, and joint elements are not needed between adjacent triangular elements. Therefore, there is no need to introduce a large virtual heat exchange coefficient between the adjacent elements. The heat conduction calculation can adopt a large time step, which significantly improves the calculation efficiency. Therefore, the model in this paper overcomes the limitations of previous models in the papers (Yan and Jiao, 2019; Yan et al., 2019) as stated in the introduction.

3.2. Calculation of heat conduction during crack propagation

Quasi-brittle material may generate thermal stress upon temperature change. If the thermal stress is large enough, the crack may initiate and propagate in the quasi-brittle material, i.e. thermal cracking. Thermal cracking will change the original mesh of thermal calculation. The node sharing relationship between the adjacent triangular elements at the crack tip will be updated. Correspondingly, the way of heat flux calculation for nodes located on the newly generated cracks will also be updated.

As shown in Fig. 6, when a new crack 1–2 is generated, the node sharing relationship on both sides of crack 1–2 is updated. The original node 1 becomes two nodes 1 and 1′, and the originally connected triangular elements Δ1,2,3 and Δ1,5,2 in Fig. 3(b) are separated to form elements Δ1,2,3 and Δ1′,5,2. These two triangular elements exchange heat through crack 1–2. Therefore, the heat flux into node 1 from Δ1,5,2 ($Q_{\Delta 1,5,2\to 1}$) should be replaced by $Q_{\Delta 1',5,2\to 1}$. At the same time, the heat flux from Δ6,9,5 into node 1 is changed to node 1′, $Q_{\Delta 6,9,5\to 1'}$. Therefore, the total heat flux into node 1 is updated from Eq. (17) to the following

$$Q_{total\to 1} = Q_{\Delta 1,2,3\to 1} + Q_{\Delta 1,3,4\to 1} + Q_{\Delta 1',5,2\to 1} + Q_{\Delta 6,10,7\to 1} \tag{16}$$

where $Q_{\Delta 1',5,2\to 1} = \frac{1}{2}h_c(T_{1'} - T_1)L_{1',2}$, $L_{1',2}$ is the length of edge 1′-2, and $T_{1'}$ is the temperature of node 1′.

Similarly, the total heat flux into the newly generated node 1′ is given by

$$Q_{total\to 1'} = Q_{\Delta 1',2,5\to 1'} + Q_{\Delta 1,2,3\to 1'} + Q_{\Delta 6,9,5\to 1'} \tag{17}$$

where $Q_{\Delta 1,2,3\to 1'} = \frac{1}{2}h_c(T_1 - T_{1'})L_{1,2}$, and $Q_{\Delta 6,9,5\to 1'} = \frac{1}{2}h_c(T_6 - T_{1'})L_{6,5}$.

Finally, the temperature of node 1 and 1′ is still updated according to Eq. (14). The temperature of other nodes on the cracks can be updated in the same way.

To this end, the model can dynamically consider the temperature evolution in the discontinuous medium during crack initiation and propagation, and consider the hindering effect of crack on heat con-

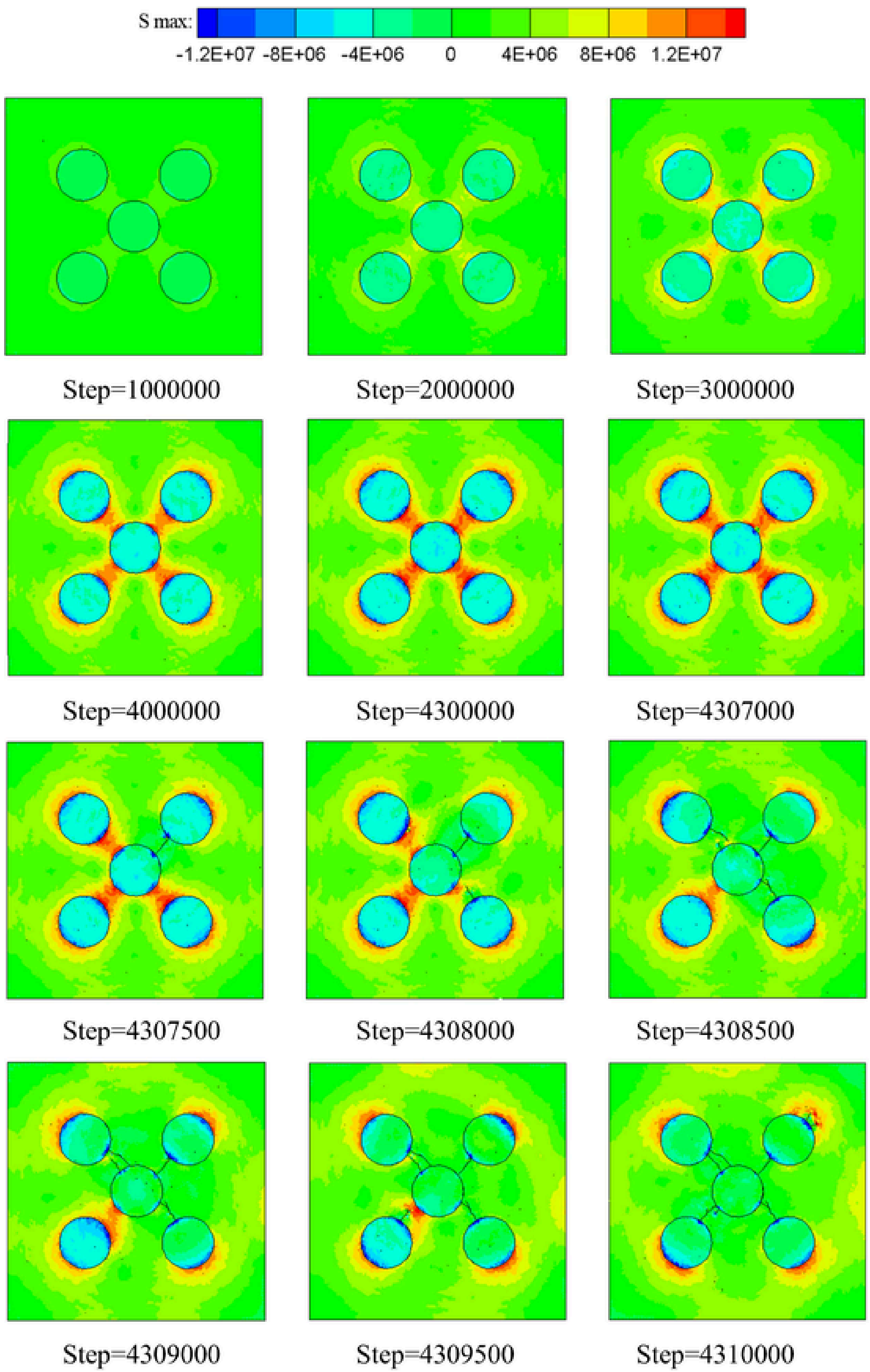

Fig. 23. Evolution of the maximum principal stress during heating.

duction, as well as the discontinuity of the temperature on both sides of the crack.

An explicit algorithm is used in the continuous-discontinuous heat conduction model. To ensure the stability of the numerical calculation,

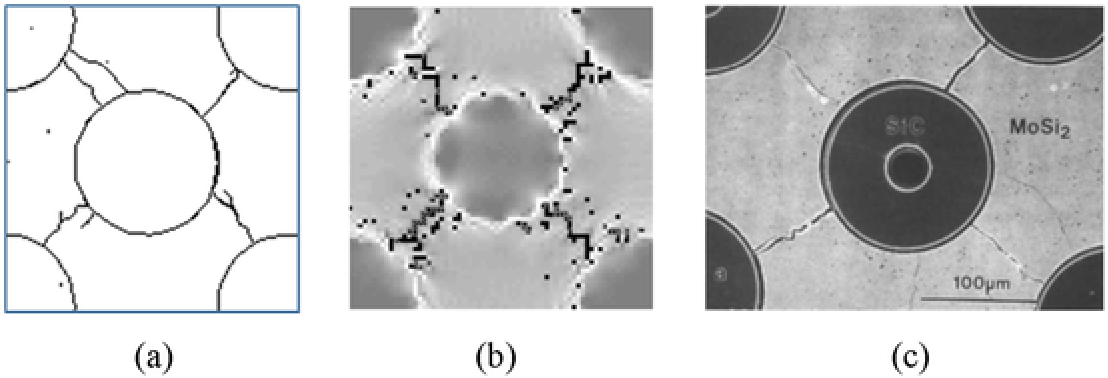

Fig. 24. **(a)** FDEM simulation result, (b) RFPA simulation result (Tang et al., 2007), (c) Laboratory experiment result (Lu et al., 1991).

the thermal time step size needs to be smaller than the critical time step size, as defined below (Itasca, 2005)

$$\Delta t_c = \frac{(L_e)^2}{4\kappa\left[1+\frac{h_c L_e}{2k}\right]} \tag{18}$$

where, L_e is the minimum element size, and κ is the thermal diffusion coefficient ($\kappa = k/\rho C_p$ when $k_x = k_y = k$).

It should be pointed out that Eq. (18) is the critical time step for heat conduction calculation, that is, the critical thermal time step. There is also a critical time step for mechanical calculation, which needs to meet two conditions: one is to satisfy the stability of the deformation calculation of the triangular element and joint element, and the other is to ensure the stability of the contact calculation between triangular elements. Among them, the critical time step that satisfies the stability of deformation calculation is

$$\Delta t_d = \min\left(\sqrt{\frac{m_i}{\max(E, p_f)}}\right) = \min\left(\sqrt{\frac{m_i}{p_f}}\right) \tag{19}$$

where i is the number of the triangular element, m_i is the mass of the triangular element, and E is elastic modulus, p_f is the penalty parameter of the joint element, and $p_f > E$ is used in deriving the above equation.

The critical time step that satisfies the stability of the contact calculation between triangular elements is

$$\Delta t_c = \min\left(\sqrt{\frac{m_i}{p_n}}\right) \tag{20}$$

where p_n is contact penalty of the triangular element in the system. Finally, the critical time step for FDEM mechanical calculation is

$$\Delta t_m = \min(\Delta t_d, \Delta t_c) \tag{21}$$

4. Thermo-mechanical coupling

A temperature change in a constrained or heterogeneous or anisotropic solid medium may induce thermal stress and even thermal cracking, which in return affects heat transfer in the medium. Therefore, the process of heat conduction is affected by the initiation and propagation of cracks. The coupled thermo-mechanical model can consider the thermal stress, thermal cracking, as well as hindrance effect of crack on the heat conduction.

4.1. Calculation of thermal stress

The thermal stress caused by the temperature change $\Delta\sigma_{ij}$ is given by

$$\Delta\sigma_{ij} = -\delta_{ij} 3K^*\alpha\Delta T \tag{22}$$

where, α is the thermal expansion coefficient, $K^* = K$ in the plane strain problem, and $K^* = 6KG/(3K+4G)$ in the plane stress problem (K is the bulk modulus, G is the shear modulus).

Next, the thermal stress, as a body load, is applied to triangular elements according to Eq. (22). The equivalent nodal force caused by the body load can be written as

$$f^{(m)} = -\frac{1}{2}\delta_{ij} 3K^*\alpha\Delta T n_j^{(m)} L^{(m)} \tag{23}$$

where, L is the length of the opposite edge of the node m in the triangular element and n_j is the outward normal unit vector of the edge.

Then, the mechanical calculation of FDEM is carried out by taking the above nodal force as input load. In this way, the thermo-mechanical coupling and thermal cracking problems can be modeled and analyzed.

4.2. Thermo-mechanical coupling process

The process of the entire thermo-mechanical coupling is divided into three steps. First, the temperature evolution of the problem domain is calculated according to heat transfer. Second, the thermal stress in all triangular elements is calculated according to the temperature evolution. In the end, the thermal stress is prescribed to the problem domain to perform FDEM mechanical calculation. The node sharing relationship of the adjacent triangular elements at the newly generated crack is updated, which is used as the mesh input for the next time step to calculate the heat transfer. Thus, the hindrance effect of crack on heat conduction is considered. According to the above three-step cycle, the thermo-mechanical coupling and thermal cracking can be modeled and analyzed. The calculation flowchart is shown in Fig. 7.

5. Numerical examples

5.1. 2D transient heat conduction problem in continuum

A classic heat conduction example (Bobaru and Duangpanya, 2010) is used here to validate the continuous-discontinuous heat conduction model. As shown in Fig. 8, a homogeneous rectangle with a length L = 10 cm and a width W = 5 cm has a fixed temperature T_R = 100 °C on the right-hand boundary. The initial temperature of the left part of the rectangular domain (x∈(0, L/2)) is T_{il} = 60 °C while in the right part of the domain (x∈(L/2, L)) T_{ir} = 0 °C. The governing equation is as follows

$$\frac{\partial T(x,t)}{\partial t} = \alpha\frac{\partial^2 T(x,t)}{\partial x^2}, \quad 0<x<\infty, 0\leqslant t<\infty \tag{24}$$

$$T(L,t) = T_R = 100°C, \quad 0\leqslant t<\infty \tag{25}$$

$$T(x,0) = \begin{cases} T_{il} = 60\ ^\circ C, & 0 < x < \frac{L}{2} \\ T_{ir} = 0\ ^\circ C, & \frac{L}{2} < x < L \end{cases} \tag{26}$$

where α is the thermal diffusion coefficient, and $\alpha = \frac{k}{\rho C_p}$, k is the thermal conductivity, ρ is the density, C_p is the specific heat, L is the length of the model, and t is time. The analytical solution is given by

$$T(x,t) = T_R + \frac{4}{\pi}\sum_{n=1,3,5,\dots}^{\infty}\frac{1}{n}\left[\left(T_{il}-T_R\right)\sin\frac{n\pi}{4} + \left(T_{ir}-T_R\right)\left(\sin\frac{n\pi}{2} - \sin\frac{n\pi}{4}\right)\right]\cos\frac{n\pi x}{2L}e^{-\alpha t\left(\frac{n^2\pi^2}{4L^2}\right)} \tag{27}$$

The calculation parameters are as follows: k is 5 W/m·°C, ρ is 1500 kg/m^3, C_p is 29 J/kg·°C. The entire model is discretized into 3537 triangular elements, and the element size ΔL is 2 mm.

Fig. 9 compares the numerical solution and analytic solution along the monitoring line (see Fig. 8) at t = 0.1 s, 4 s, 8 s. The calculation results of heat conduction for continuum are naturally consistent with analytical solutions because the continuous-discontinuous heat conduction model does not need to consider the hindering effect of unbroken joint elements on heat conduction.

To study the effect of mesh size on heat transfer, here we run the heat conduction example with three different mesh sizes (L_e = 2, 4, 6, 8 mm), and we set a monitoring line at y = 2.5 cm.

Fig. 10 shows the temperature distribution when L_e is 2 mm, 4 mm, 6 mm, and 8 mm at t = 4 s. Fig. 11 shows the comparison between the numerical results and the analytical solution at the monitoring line (y = 2.5 cm). It can be seen that by reducing the mesh size, the numerical result converges to the analytical solution.

5.2. Heat conduction in a medium with a single crack

Heat conduction in a rectangular domain with a single crack (Ezekoye, 2016; Sun et al., 2020; Wu et al., 2019) is conducted to verify the 2D continuous-discontinuous heat conduction model and investigate the hindrance effect of the crack on heat conduction. As shown in Fig. 12, a square with a length (2L) of 6 m and a width (W) of 3 m contains a crack located at × = 3 m. Temperatures at the left and right boundary are fixed as T_0 = 100 °C and T_1 = 0 °C respectively. The initial temperature in the square is 0 °C. We derived a new analytical solution applicable for any value of L, as

$$\begin{cases} T = \frac{2-x/L+(k/L)/h_c}{2+(k/L)/h_c}\left(T_0-T_1\right), & for\ x\in(0,L) \\ T = \frac{2-x/L}{2+(k/L)/h_c}\left(T_0-T_1\right), & for\ x\in(L,2L) \end{cases} \tag{28}$$

where k is the thermal conductivity and h_c the heat exchange coefficient of cracks.

Other relevant parameters are given as follows: k is 1.5 W/ (m·°C), and h_c are 50 W/ (m^2·°C), 5 W/ (m^2·°C), 0.5 W/(m^2·°C), 0.05 W/(m^2·°C), 0.005 W/(m^2·°C), 0.0005 W/(m^2·°C).

As shown in Fig. 13, the numerical solution agrees well with the analytical solution. When the ratio of $h_c/(k/L)$ is smaller than 0.01 (i.e., h_c < 0.005 W/(m^2·°C)), the crack almost blocks the heat transfer in the plate, and the temperature jump on two sides of the crack is almost 100 °C. As the value of $h_c/(k/L)$ increases, the temperature discontinuity across the crack decreases. When the value of $h_c/(k/L)$ is larger than 100 (i.e., h_c > 50 W(m^2·°C)), the temperature discontinuity is close to zero, and the crack has very little hindrance to heat conduction.

5.3. Heat conduction in a medium with multiple cracks

We design another example of heat transfer in a medium with multiple cracks. The effect of the heat exchange coefficient of cracks, h_c, is investigated to prove the broad applicability of the model in this paper. A rectangle with a length (L) of 10 cm and a width (W) of 5 cm contains various forms of cracks as classified in the literature (Bobaru and Duangpanya, 2012; Chen and Chang, 1994), including penetrating, no penetrating, crossing, parallel, and inclined cracks. To facilitate observation and analysis, we set a monitoring line at y = 2.5 cm shown in Fig. 14. The temperatures at the left and right boundary are fixed at 100 °C and 0 °C respectively. The initial temperature within the square domain is 0 °C.

Other relevant parameters are as follows: k is 1.5 W/(m·°C), and h_c are set as 7500 W/(m^2·°C), 750 W/(m^2·°C), 75 W/(m^2·°C), 7.5 W/(m^2·°C), 0.75 W/(m^2·°C), and 0 W/(m^2·°C).

When heat conduction reaches a steady state, a comparison between the numerical solution and the analytical solution is shown in Fig. 15. The difference in the heat exchange coefficient of cracks has a great effect on heat transfer. Taking h_c = 0 W/(m^2·°C) as an example, the crack completely blocks the heat transfer from left to right. The temperature change across point A on the monitoring line is almost 100 °C because A is located on a penetrating crack. As the value of h_c/(k/L) increases, the temperature change across point A decreases. When h_c = 7500 W/(m^2·°C), the hindering effect of cracks on heat conduction becomes negligible, and the temperature variation appears to be continuous.

Fig. 16 shows the temperature distribution of the model when the heat conduction reaches a steady state. For h_c = 0 W/(m^2·°C), five intersecting cracks form a network penetrating through the width dimension of the domain, which completely blocks the heat transfer from the left to the right side through the cracks. The temperature difference between the two sides of the connected crack is very obvious. As the value of h_c increases, the temperature change across the cracks decreases. When h_c = 7500 W/(m^2·°C), the cracks have almost no hindrance to heat conduction. There is only a slight difference in temperature between the two sides of the cracks. It can be seen that the continuous-discontinuous heat conduction model can well consider the hindering effect of cracks on heat conduction and the temperature discontinuity on both sides of the crack.

5.4. Stress intensity factor

This example is a single edge notched (SEN) plate under uniform tensile stress P = 1 MPa as shown in the Fig. 17. The size of the plate is 0.4 m in length (2L), 0.1 m in width (W), and the crack length (a) is 0.02 m. Here we solve the stress intensity factor (SIF) of the mode I crack using FDEM and compared it with the following analytical solution (Tada et al., 1973)

$$K_I^{ref} = P\sqrt{\pi a}\sqrt{\frac{2W}{\pi a}\tan\frac{\pi a}{2W}}\frac{0.752 + 2.02\frac{a}{W} + 0.37\left(1-\sin\frac{\pi a}{2W}\right)}{\cos\frac{\pi a}{2W}} \tag{29}$$

As shown in Fig. 18, stress concentration appears at the crack tip under tensile stress P. Using the displacement distribution around the crack obtained by FDEM, the stress intensity factor at the crack tip is calculated according to an extrapolation method (Chen and Kuang, 1992). The relative error of the numerical result (K_I = 0.3508 MPa·$\sqrt{\text{m}}$) to the analytical solution (K_I^{ref} = 0.3784 MPa·$\sqrt{\text{m}}$) is 7.3%, which falls into a reasonable range. Of course, the numerical accuracy can be further improved by mesh refinement. This example demonstrated that FDEM is effective and accurate to simulate stress distribution around the crack tip.

6. Thermal cracking examples

6.1. Cracking induced by temperature gradient

To verify the applicability of the coupled thermo-mechanical model to model the thermal cracking problem, an experiment of reference (Jansen et al., 1993) is simulated. A square plate with an edge length (L) of 15 cm and a center borehole with a diameter (d) of 1 cm are shown in Fig. 19. The top and bottom boundaries are fixed in the normal direction, and free in the tangential direction. The initial temperature in the square and its four boundaries is 20 °C, and temperature at the borehole wall gradually increases from 20 °C to 200 °C (0.0001 °C/step, step = 0–1.8e6). Four side boundaries of the plate are insulated.

The mechanical and thermal parameters are obtained from the experimental data of LDB granite in reference (Martin, 1993). Some important parameters are as follows: the Poisson's ratio ν is 0.25, the density ρ is 2600 m/s^2, the elastic modulus E is 69 GPa, internal friction angle φ is 59°, the tensile strength f_t is 9 MPa, the thermal expansion coefficient $\alpha$is 1×10^{-5}/°C, and the thermal conductivity k is 1.2 W/(m. °C).

As shown in Fig. 19, as the temperature around the boundary of the central hole increases, a tensile crack is first initialized at the bottom edge of the plate, followed by a second crack at the top of the plate. They both propagate towards the center hole. This failure characteristic is consistent with that in the literature (Jansen et al., 1993; Zhao, 2015) shown in Fig. 20.

Fig. 21 is the distribution of the maximum principal stress of the model at different time steps. As the boundary temperature of the central hole increases, heat transfers outward from the boundary of the hole. The rock around the hole has experienced a large expansion deformation due to temperature increase, while expansion deformation is small for the rock at the external boundary because of a less increase in the temperature. Therefore, the external boundary exerts a restraining effect on the internal expansion deformation. The concentration of compressive stress (negative values) is observed around the center hole. On the other hand, tensile stress/deformation (positive values) is produced at the external boundary due to the volumetric expansion of the interior. Taking step = 100,000–1,572,000 as an example, as the temperature increases, the concentration of tensile stress occurs first at the center of the top and bottom boundaries. As shown in Fig. 21, since the top and bottom boundaries are fixed in the normal direction and the left and right boundaries are free, tensile deformation can only occur along the horizontal direction. When the tensile stress of the specimen reaches its tensile strength, cracks occur at the center of the bottom boundaries, as shown in step = 1,561,000. A second crack is initiated at the top boundary at step 1,564,000. Both cracks propagate perpendicular to the edges towards the center hole. At the same time, the tensile stress concentration at cracks is released. The new tensile stress concentration appears at the tip of cracks and moves inward as cracks extend. Eventually, two vertical cracks penetrate through the center hole at step = 1,572,000. This example well illustrates the applicability of the coupled thermo-mechanical model in modeling the thermal cracking process in quasi-brittle materials.

Fig. 22 shows the evolution of strain energy (E_s), kinetic energy (E_k), and the number of broken joint elements during the heating process. As the temperature of borehole continues to increase, the strain energy increases until cracks are generated, which is represented by sharp increase of broken joint elements within 11,000 steps. After fracturing, 65% strain energy is released and part of it (18%) is converted into kinetic energy when the broken pieces move apart in two horizontal directions. As the temperature of the borehole boundary continues to increase, the strain energy increases until the borehole boundary temperature remains constant. After that, the strain energy and kinetic energy no longer change.

6.2. Cracking induced by thermal mismatch

In addition to the thermal stress induced by a temperature gradient, the thermal mismatch can also generate thermal stress in quasi-brittle materials. As an example, we simulate a thermal cracking problem (Tang et al., 2007) caused by the thermal mismatch. Within a 1 m × 1 m square domain, five circular inclusions with a radius of 0.1 m are embedded in the matrix. The entire sample is subjected to a uniform temperature increase at a rate of 0.0001 °C/step, changing from 20 to 500 °C. The material parameters used in the simulation are shown in Table 1.

Fig. 23 shows the evolution of the maximum principal stress when the model at different calculation steps. We stipulate that the tensile stress is positive and the compressive stress is negative. Since the thermal expansion coefficient of the inclusions is greater than that of the matrix, the matrix is in a compressive state along the radial direction while in a tensile state along the circumferential direction in the areas surrounding the embedded particles, as shown in Fig. 23 (the maximum principal stress is tension). In the areas connecting the center of the middle particle and its surrounding particles, the circumferential tensile stress and radial compressive stress become the largest due to the superposition of the stress field around these particles. On the other hand, the embedded particles are in a compressive state in both radial and circumferential directions. When the tensile stress of the matrix reaches its tensile strength, cracks begin to generate in the matrix, see Fig. 23 step 4,307,000 for the initiation of a tensile crack at the matrix/inclusion interface around the center particle. The crack extends along the radial direction and connects the neighboring particle at step 4307500. Finally, as the temperature rises, more cracks are generated and propagate to connect the middle particle and the surrounding particles, as shown in steps 4,307,500–4,310,000. It is interesting to note that, the tensile stress in the matrix is released around the cracked area, while the particles are still in a compressive state due to the persistence of the compressive interactive force between the matrix and inclusions in the radial direction. The crack pattern obtained in this study is in good agreement with rock failure process analysis (RFPA) (Tang et al., 2007). A similar phenomenon has also been observed in an experimental study (LU et al., 1991), as shown in Fig. 24(c). In summary, the difference in thermal expansion coefficients of the matrix and the particles produce significant tensile stress concentration at the interface between the matrix and the inclusion. Finally, radial cracks are formed in the matrix connecting the center particle and four surrounding particles, resulting in a redistributed stress field. The FDEM simulation result demonstrates that the coupled thermo-mechanical model in this paper is effective in simulating thermal cracking problems.

7. Conclusions

In this study, we propose a new 2D continuous-discontinuous heat conduction model based on the Finite-Discrete Element Framework. The new model has significant advantages over previous models, such that there is no need to introduce artificial joint elements in the continuum domain of heat conduction. Therefore, the computational time step of thermal calculation can be significantly improved compared with previously developed heat conduction models based on FDEM. Besides, the continuous-discontinuous heat conduction model can dynamically update the node sharing relationship between the adjacent triangular elements along the crack during crack propagation. Therefore, the hindrance of crack on heat conduction and the temperature discontinuity on both sides of the crack can be well considered. The above continuous-discontinuous heat conduction model is combined with the FDEM mechanical calculation to set up a coupled thermo-mechanical model, which has demonstrated excellent capability in simulating thermal cracking of quasi-brittle materials.

In addition to FDEM, the 2D continuous-discontinuous heat conduction model can be combined with other numerical methods, such as FEM, DEM, and DDA, to construct corresponding thermo-mechanical models to solve heat transfer, thermal cracking problems in quasi-brittle materials.

It should be pointed out that the coupled thermo-mechanical model is based on FDEM. Therefore, when simulating thermal cracking, the crack propagation proceeds along the element boundary and thus is mesh-dependent. However, the mesh dependence of crack propagation gradually decreases as the mesh size decreases. In FDEM, the low-order triangular element is used for discretization, which results in a loss of calculation accuracy. Furthermore, input parameters for FDEM, such as elastic modulus, Poisson's ratio, etc. can be directly obtained. However, parameters of the fracture model are relatively difficult to calibrate. The above issues still need to be carefully studied in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[Ahmed et al., 2018] Ahmed MF, Waqas U, Arshad M, Rogers JD Effect of heat treatment on dynamic properties of selected rock types taken from the Salt Range in Pakistan. Arab. J. Geosci. 2018;11(22):728.
[Amin et al., 2009] Amin MN, Kim J-S, Lee Y, Kim J-K Simulation of the thermal stress in mass concrete using a thermal stress measuring device. Cem. Concr. Res. 2009;39(3):154–64.
[An et al., 2018] An Z, Yu T, Bui TQ, Wang C, Trinh NA Implementation of isogeometric boundary element method for 2-D steady heat transfer analysis. Adv. Eng. Softw. 2018;116:36–49.
[Areias and Belytschko, 2005] Areias PMA, Belytschko T Non-linear analysis of shells with arbitrary evolving cracks using XFEM. Int. J. Numer. Methods Eng. 2005;62(3):384–415.
[Bazazzadeh et al., 2020] Bazazzadeh S, Mossaiby F, Shojaei A An adaptive thermo-mechanical peridynamic model for fracture analysis in ceramics. Eng. Fract. Mech. 2020;223:106708.
[Belytschko and Black, 1999] Belytschko T, Black T Elastic crack growth in finite elements with minimal remeshing. Int. J. Numer. Methods Eng. 1999;45:601–20.
[Benabou et al., 2013] Benabou L, Sun Z, Dahoo PR A thermo-mechanical cohesive zone model for solder joint lifetime prediction. Int. J. Fatigue 2013;49:18–30.
[Bobaru and Duangpanya, 2010] Bobaru F, Duangpanya M The peridynamic formulation for transient heat conduction. Int. J. Heat Mass Transf. 2010;53(19–20):4047–59.
[Bobaru and Duangpanya, 2012] Bobaru F, Duangpanya M A peridynamic formulation for transient heat conduction in bodies with evolving discontinuities. J. Comput. Phys. 2012;231(7):2764–85.
[Borden et al., 2018] Borden MJ, Hughes TJR, Landis CM, Anvari A, Lee IJ Phase-field formulation for ductile fracture, advances in computational plasticity. Comput. Methods Appl. Sci. 2018;45–70.
[Bou Jaoude et al., 2018] Bou Jaoude I, Novakowski K, Kueper B Identifying and assessing key parameters controlling heat transport in discrete rock fractures. Geothermics 2018;75:93–104.
[Caggiano and Etse, 2015] Caggiano A, Etse G Coupled thermo–mechanical interface model for concrete failure analysis under high temperature. Comput. Meth. Appl. Mech. Eng. 2015;289:498–516.
[Carslaw and Jaeger, 1959] Carslaw HS, Jaeger JC Conduction of heat in solids. Oxford: Oxford University; 1959.
[Chen and Kuang, 1992] Chen LS, Kuang JH A displacement extrapolation method for determining the stress intensity factors along flaw border. Int. J. Fract. 1992;57:51–8.
[Chen and Chang, 1994] Chen W-H, Chang C-L Heat conduction analysis of a plate with multiple insulated cracks by the finite element alternating method. Int. J. Solids Struct. 1994;31(10):1343–55.
[Choo et al., 2012] Choo J, et al. Stress-induced evolution of anisotropic thermal conductivity of dry granular materials. Acta Geotech. 2012;8(1):91–106.
[Chu et al., 2017] Chu D, Li X, Liu Z Study the dynamic crack path in brittle material under thermal shock loading by phase field modeling. Int. J. Fract. 2017;208(1–2):115–30.
[Committee2007] Committee, A.C.I., 2007. Report on thermal and volume change effects on cracking of mass concrete.
[Cui et al., 2019] Cui Y-Q, Yu Z-Q, Lan P A novel method of thermo-mechanical coupled analysis based on the unified description. Mech. Mach. Theory 2019;134:376–92.
[Deng and Liu, 2020] Deng P, Liu Q Influence of the softening stress path on crack development around underground excavations: Insights from 2D-FDEM modelling. Comput. Geotech. 2020;117:103239.
[Ekneligoda and Marshall, 2018] Ekneligoda TC, Marshall AM A coupled thermal-mechanical numerical model of underground coal gasification (UCG) including spontaneous coal combustion and its effects. Int. J. Coal Geol. 2018;199:31–8.
[Ezekoye, 2016] Ezekoye OA Conduction of Heat in Solids. SFPE Handbook of Fire Protection Engineering 2016;25–52.
[Fang et al., 2020] Fang W, An Z, Yu T, Bui TQ Isogeometric boundary element analysis for two-dimensional thermoelasticity with variable temperature. Eng. Anal. Bound. Elem. 2020;110:80–94.
[Fu et al., 2017] Fu Y, Wang Z, Ren F, Wang D Numerical model of thermo-mechanical coupling for the tensile failure process of brittle materials. AIP Adv. 2017;7(10):105023.
[Fukuda et al., 2019] Fukuda D, et al. Development of a GPGPU-parallelized hybrid finite-discrete element method for modeling rock fracture. Int. J. Numerical Anal. Methods Geomech. 2019.
[Gao et al., 2015] Gao X-W, Zheng B-J, Yang K, Zhang C Radial integration BEM for dynamic coupled thermoelastic analysis under thermal shock loading. Comput. Struct. 2015;158:140–7.
[Gautam et al., 2018] Gautam PK, Verma AK, Jha MK, Sharma P, Singh TN Effect of high temperature on physical and mechanical properties of Jalore granite. J. Appl. Geophys. 2018;159:460–74.
[Heuze, 1983] Heuze FE High-temperature mechanical, physical and Thermal properties of granitic rocks — a review. Int. J. Rock Mech. Mining Sci. Geomech. Abstracts 1983;20(1):3–10.
[Itasca2005] Itasca, 2005. Code UDEC. User's Guide. Itasca Consulting Group Inc, Minnesota, USA.
[Jansen et al., 1993] Jansen DP, Carlson SR, Young RP, Hutchins DA Ultrasonic imaging and acoustic emission monitoring of thermally induced microcracks in Lac du Bonnet granite. J. Geophys. Res. Solid Earth 1993;98(B12):22231–43.
[Jiangetal2020] Jiang, W., Spencer, B.W., Dolbow, J.E., 2020. Ceramic nuclear fuel fracture modeling with the extended finite element method. Eng. Fract. Mech., 223.
[Jiao et al., 2015] Jiao Y-Y, et al. A coupled thermo-mechanical discontinuum model for simulating rock cracking induced by temperature stresses. Comput. Geotech. 2015;67:142–9.
[Kidane et al., 2010] Kidane A, Chalivendra VB, Shukla A Thermo-mechanical stress fields and strain energy associated with a mixed-mode propagating crack. Acta Mech. 2010;215(1–4):57–69.
[Kim et al., 2011] Kim J-S, Kwon S-K, Sanchez M, Cho G-C Geological storage of high level nuclear waste. KSCE J. Civ. Eng. 2011;15(4):721–37.
[Kumari et al., 2019] Kumari WGP, et al. An experimental study on tensile characteristics of granite rocks exposed to different high-temperature treatments. Geomech. Geophys. Geo-energ. Geo-resour. 2019;5(1):47–64.
[Kumari et al., 2017] Kumari WGP, Ranjith PG, Perera MSA, Chen BK, Abdulagatov IM Temperature-dependent mechanical behaviour of Australian Strathbogie granite with different cooling treatments. Eng. Geol. 2017;229:31–44.
[Lei et al., 2020] Lei Z, Rougier E, Euser B, Munjiza A A smooth contact algorithm for the combined finite discrete element method. Computational Particle Mech. 2020;7(5):807–21.
[Lei et al., 2019] Lei Z, Rougier E, Munjiza A, Viswanathan H, Knight EE Simulation of discrete cracks driven by nearly incompressible fluid via 2D combined finite-discrete element method. Int. J. Numer. Anal. Meth. Geomech. 2019;43(9):1724–43.
[Li and Shirvan, 2021] Li W, Shirvan K Multiphysics phase-field modeling of quasi-static cracking in urania ceramic nuclear fuel. Acta Mech. 2021;47(1):793–810.
[Li et al., 2020] Li XF, et al. Investigating the crack initiation and propagation mechanism in brittle rocks using grain-based finite-discrete element method. Int. J. Rock Mech. Min. Sci. 2020;127.
[Lietal2021] Li, X.F., Li, H.B., Zhao, J., 2021. Transgranular fracturing of crystalline rocks and its influence on rock strengths: Insights

from a grain-scale continuum–discontinuum approach. Computer Methods in Applied Mechanics and Engineering, 373.
[Lisjak et al., 2013] Lisjak A, Liu Q, Zhao Q, Mahabadi OK, Grasselli G Numerical simulation of acoustic emission in brittle rocks by two-dimensional finite-discrete element analysis. Geophys. J. Int. 2013;195(1):423–43.
[Liu et al., 2019] Liu H, Zhang K, Shao S, Ranjith PG Numerical investigation on the mechanical properties of Australian Strathbogie granite under different temperatures using discrete element method. Rock Mech. Rock Eng. 2019;52(10):3719–35.
[Liuetal2020a] Liu, J. et al., 2020a. Micro-cracking behavior of shale matrix during thermal recovery: Insights from phase-field modeling. Eng. Fract. Mech., 239.
[Liu et al., 2020b] Liu L, et al. Dual-damage constitutive model to define thermal damage in rock. Int. J. Rock Mech. Min. Sci. 2020b;126:104185.
[Lu et al., 1991] Lu TC, et al. Matrix cracking in intermetallic composites caused by thermal expansion mismatch. Acta Metall. Mater. 1991;39(8):1883–90.
[Mahanta et al., 2016] Mahanta B, Singh TN, Ranjith PG Influence of thermal treatment on mode I fracture toughness of certain Indian rocks. Eng. Geol. 2016;210:103–14.
[Martin1993] Martin, C.D., 1993. The strength of massive Lac du Bonnet granite around underground openings.
[Meguid and Hu, 1999] Meguid SA, Hu GD A new finite element for treating plane thermomechanical heterogeneous solids. Int. J. Numer. Methods Eng. 1999;44(4):567–85.
[Miehe et al., 2015a] Miehe C, Hofacker M, Schänzel LM, Aldakheel F Phase field modeling of fracture in multi-physics problems. Part II. Coupled brittle-to-ductile failure criteria and crack propagation in thermo-elastic–plastic solids. Comput. Meth. Appl. Mech. Eng. 2015a;294:486–522.
[Miehe et al., 2015b] Miehe C, Schänzel L-M, Ulmer H Phase field modeling of fracture in multi-physics problems. Part I. Balance of crack surface and failure criteria for brittle crack propagation in thermo-elastic solids. Comput. Meth. Appl. Mech. Eng. 2015b; 294:449–85.
[Moës et al., 2017] Moës N, Dolbow JE, Sukumar N Extended finite element methods. Encyclopedia of Computational Mech. Second Edition 2017;1–21.
[Munjiza, 2004] Munjiza A The Combined Finite-discrete Element Method. John Wiley and Sons Ltd.; 2004.
[Munjiza and Andrews, 1998] Munjiza A, Andrews KRF Contact detection algorithm for bodies of similar size. Int. J. Numer. Methods Eng. 1998;43:131–49.
[Munjiza and Andrews, 2000] Munjiza A, Andrews KRF Penalty function method for combined finite–discrete element systems comprising large number of separate bodies. Int. J. Numer. Methods Eng. 2000;49 (11):1377–96.
[Munjiza et al., 1999a] Munjiza A, Andrews KRF, White JK Combined single and smeared crack model in combined finite-discrete element analysis. Int. J. Numer. Methods Eng. 1999a;44:41–57.
[Munjiza and John, 2002] Munjiza A, John NWM Mesh size sensitivity of the combined FDEM fracture and fragmentation algorithms. Eng. Fract. Mech. 2002;69(2):281–95.
[Munjiza et al., 1999b] Munjiza A, Latham JP, Andrews KRF Challenges of a coupled combined finite-discrete element approach to explosive induced rock fragmentation. Fragblast 1999b;3(3):237–50.
[Munjiza et al., 1995] Munjiza A, Owen DRJ, Bicanic N A combined finite-discrete element method in transient dynamics of fracturing solids. Eng. Comput. 1995;12:145–74.
[Najafi et al., 2014] Najafi M, Jalali SME, KhaloKakaie R Thermal–mechanical–numerical analysis of stress distribution in the vicinity of underground coal gasification (UCG) panels. Int. J. Coal Geol. 2014;134–135:1–16.
[Nechnech et al., 2002] Nechnech W, Meftah F, Reynouard JM An elasto-plastic damage model for plain concrete subjected to high temperatures. Eng. Struct. 2002;24:597–611.
[Ouedraogo et al., 2011] Ouedraogo E, Roosefid M, Prompt N, Deteuf C Refractory concretes uniaxial compression behaviour under high temperature testing conditions. J. Eur. Ceram. Soc. 2011; 31(15):2763–74.
[Peng et al., 2019] Peng J, Rong G, Tang Z, Sha S Microscopic characterization of microcrack development in marble after cyclic treatment with high temperature. Bull. Eng. Geol. Environ. 2019;78(8): 5965–76.
[Rodriguez et al., 2016] Rodriguez JM, Carbonell JM, Cante JC, Oliver J The particle finite element method (PFEM) in thermo-mechanical problems. Int. J. Numer. Methods Eng. 2016;107(9):733–85.
[Shou and Zhou, 2019] Shou Y, Zhou X A coupled thermomechanical nonordinary state-based peridynamics for thermally induced cracking of rocks. Fatigue Fract. Eng. Mater. Struct. 2019;43(2): 371–86.
[Sun et al., 2020] Sun L, Liu Q, Grasselli G, Tang X Simulation of thermal cracking in anisotropic shale formations using the combined finite-discrete element method. Comput. Geotech. 2020;117: 103237.
[Sun et al., 2016] Sun Q, et al. Thermal properties of sandstone after treatment at high temperature. Int. J. Rock Mech. Min. Sci. 2016;85:60–6.
[Tada et al., 1973] Tada H, Paris PC, Irwin GR The Stress Analysis of Cracks Handbook. Del Research Corp; 1973.
[Tang et al., 2007] Tang SB, Tang CA, Liang ZZ, Yu QL Numerical model for thermal cracking of concrete. Key Eng. Mater. 2007;353–358 (Pt2):941–4.
[Taylor and Bryant, 2014] Taylor J, Bryant S Quantifying thermally driven fracture geometry during CO2 storage. Energy Procedia 2014; 63:3390–404.
[Ulm et al., 1999] Ulm F-J, Coussy O, BažAnt ZP The "Chunnel" fire. I Chemoplastic softening in rapidly heated concrete. J. Eng. Mech. 1999;125(3):272–82.
[Wang et al., 2019a] Wang F, Konietzky H, Herbst M Influence of heterogeneity on thermo-mechanical behaviour of rocks. Comput. Geotech. 2019a;116:103184.
[Wang and Zhou, 2019] Wang Y-T, Zhou X-P Peridynamic simulation of thermal failure behaviors in rocks subjected to heating from boreholes. Int. J. Rock Mech. Min. Sci. 2019;117:31–48.
[Wang et al., 2018] Wang Y, Zhou X, Kou M A coupled thermo-mechanical bond-based peridynamics for simulating thermal cracking in rocks. Int. J. Fract. 2018;211(1–2):13–42.
[Wang et al., 2019b] Wang Y, Zhou X, Kou M An improved coupled thermo-mechanic bond-based peridynamic model for cracking behaviors in brittle solids subjected to thermal shocks. Eur. J. Mech. A-Solids 2019b;73:282–305.
[Wang et al., 2019c] Wang Y, Zhou X, Zhang T Size effect of thermal shock crack patterns in ceramics: Insights from a nonlocal numerical approach. Mater.: Mech; 2019c. p. 137.
[Wu et al., 2019] Wu Z, Zhou Y, Fan L A fracture aperture dependent thermal-cohesive coupled model for modelling thermal conduction in fractured rock mass. Comput. Geotech. 2019;114:103108.
[Xia, 2015] Xia M An upscale theory of thermal-mechanical coupling particle simulation for non-isothermal problems in two-dimensional quasi-static system. Eng. Comput. 2015;32(7): 2136–65.
[Xia et al., 2014] Xia M, Zhao C, Hobbs BE Particle simulation of thermally-induced rock damage with consideration of temperature-dependent elastic modulus and strength. Comput. Geotech. 2014;55:461–73.
[Yan and Jiao, 2018] Yan C, Jiao Y-Y A 2D fully coupled hydro-mechanical finite-discrete element model with real pore seepage for simulating the deformation and fracture of porous medium driven by fluid. Comput. Struct. 2018;196:311–26.
[Yan and Jiao, 2019] Yan C, Jiao Y-Y A 2D discrete heat transfer model considering the thermal resistance effect of fractures for simulating the thermal cracking of brittle materials. Acta Geotech. 2019;15(5):1303–19.
[Yan et al., 2018] Yan C, Jiao Y-Y, Zheng H A fully coupled three-dimensional hydro-mechanical finite discrete element approach with real porous seepage for simulating 3D hydraulic fracturing. Comput. Geotech. 2018;96:73–89.
[Yan et al., 2019] Yan C, Jiao YY, Zheng H A three-dimensional heat transfer and thermal cracking model considering the effect of cracks on heat transfer. Int. J. Numer. Anal. Methods Geomech. 2019;43 (10):1825–53.
[Yan and Zheng, 2016] Yan C, Zheng H A two-dimensional coupled hydro-mechanical finite-discrete model considering porous media flow for simulating hydraulic fracturing. Int. J. Rock Mech. Min. Sci. 2016;88:115–28.
[Yan and Zheng, 2017a] Yan C, Zheng H A coupled thermo-mechanical model based on the combined finite-discrete element method for simulating thermal cracking of rock. Int. J. Rock Mech. Min. Sci. 2017a;91:170–8.
[Yan and Zheng, 2017b] Yan C, Zheng H FDEM-flow3D: A 3D hydro-mechanical coupled model considering the pore seepage of rock matrix for simulating three-dimensional hydraulic fracturing. Comput. Geotech. 2017b;81:212–28.
[Yan and Zheng, 2017c] Yan C, Zheng H Three-dimensional hydromechanical model of hydraulic fracturing with arbitrarily discrete fracture networks using finite-discrete element method. Int. J. Geomech. 2017c;17(6):4016133.
[Yang et al., 2017] Yang K, Peng HF, Wang J, Xing CH, Gao XW Radial integration BEM for solving transient nonlinear heat conduction with temperature-dependent conductivity. Int. J. Heat Mass Transf. 2017;108:1551–9.
[Zhao et al., 2019] Zhao XG, et al. Thermal conductivity of thermally damaged Beishan granite under uniaxial compression. Int. J. Rock Mech. Min. Sci. 2019;115:121–36.
[Zhao, 2015] Zhao Z Thermal influence on mechanical properties of granite: a microcracking perspective. Rock Mech. Rock Eng. 2015;49(3): 747–62.